A CENTERED VOLTERRA–FREDHOLM REALIZATION OF THE RIEMANN Ξ-FUNCTION

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Abstract. Within the Hilbert–Polya program, we construct a compact integral realization of the completed zeta function directly from its positive Theta kernel, while separating this realization from the still-open problem of spectral reality. Set $\Xi(z) = \xi(\frac{1}{2} + iz)$, and let W be the positive, even Theta density whose Fourier transform is Ξ. After normalizing $p = W/\Xi(0)$ and writing $F(x) = \int_{-\infty}^{x} p(y)\,\mathrm{d}y$, we introduce on $L^2(\mathbb{R})$ the centered Volterra operator

$$(\mathcal{B}_\Theta f)(x) = \frac{i}{\sqrt{p(x)}} \left(\int_{-\infty}^{x} \sqrt{p(y)} f(y)\,\mathrm{d}y - F(x) \int_{\mathbb{R}} \sqrt{p(y)} f(y)\,\mathrm{d}y \right).$$

We prove that $\mathcal{B}_\Theta$ is Hilbert–Schmidt, obtain an explicit norm bound, and establish the regularized determinant identity

$$\det{}_2(I + z\mathcal{B}_\Theta) = \frac{\Xi(z)}{\Xi(0)}, \qquad z \in \mathbb{C}.$$

Within the natural class of scalar rank-one endpoint corrections, the CDF term is uniquely forced by requiring differentiation to remove exactly the $\sqrt{p}$ mode. Thus the construction is canonical for the underlying probability law rather than an ad hoc determinant reduction. Consequently the nonzero spectrum of $\mathcal{B}_\Theta$ consists, with algebraic multiplicity, of the reciprocal negatives of the zeros of Ξ. We also construct a closed first-order operator $A = p^{-1/2}(-i\,d/dx)p^{1/2}$ and prove directly that $\mathcal{B}_\Theta = A^\dagger$, its Moore–Penrose inverse. The associated block supercharge gives a one-dimensional Witten-type factorization, and A is Fredholm of index -1. For an even probability density, A and $\mathcal{B}_\Theta$ are PT-symmetric and parity-odd. The determinant formula is first proved for an arbitrary density on a finite interval and is then extended to an explicit class of whole-line densities by Hilbert–Schmidt convergence. This yields, in particular, trace–cumulant identities for the associated operators. The construction is unconditional; it does not prove the Riemann hypothesis. Rather, the hypothesis becomes the precise additional assertion that the nonzero spectrum of this concrete compact PT-symmetric operator is real.

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1. Introduction

Write

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \qquad \Xi(z) = \xi\left(\frac{1}{2}+iz\right). \tag{1.1}$$

The Riemann hypothesis (RH) is equivalent to the assertion that every zero of the even entire function Ξ is real. A traditional Hilbert–Polya program therefore asks for an operator whose spectral points are the zeros of Ξ and whose reality follows from self-adjointness. Connes's noncommutative geometric realization [4] is a foundational example of a broader spectral interpretation: the critical zeros occur as an absorption spectrum for the idele-class action on the noncommutative space of Adele classes, while the explicit formulas are interpreted through a trace formula. Recent work of Hedenmalm [6] gives a Theta-based boundary-value formulation of the zeros, including complex zeros, and isolates the search for a suitable inner product. The present paper takes a different but compatible step: it turns the same positive Theta density into an explicit compact integral operator and computes its regularized Fredholm determinant exactly.

Our main object is not obtained from a table of zeros. It is constructed directly from the classical Riemann Fourier kernel. Let p be the normalized kernel, let F be its cumulative distribution function, and define

$$(\mathcal{B}_\Theta f)(x) = \frac{i}{\sqrt{p(x)}}\left[\int_{-\infty}^{x}\sqrt{p(y)}f(y)\,\mathrm{d}y - F(x)\int_{\mathbb{R}}\sqrt{p(y)}f(y)\,\mathrm{d}y\right]. \tag{1.2}$$

The first term records the ordered Volterra history $y \le x$. The rank-one second term is forced by the two endpoint conditions: it removes precisely the outgoing scalar left by the first term. More strongly, Proposition 3.2 shows that F is the unique scalar correction for which differentiation of the ordered primitive returns the orthogonal projection off the $\sqrt{p}$ mode. Thus CDF centering is not selected merely to make a determinant calculation close: it is forced by a local projection-and-endpoint principle.

It is useful to distinguish this construction from two more immediate realizations of a characteristic function. If M_x denotes multiplication by the coordinate on $L^2(\mathbb{R})$, then every probability density has the unitary matrix-coefficient representation

$$\varphi_p(z) = \left\langle e^{izM_x}\sqrt{p}, \sqrt{p}\right\rangle. \tag{1.3}$$

Although canonical, (1.3) does not turn the zeros of φ_p into spectral points of a single compact operator. At the other extreme, a diagonal or canonical-product model assembled from a prescribed zero set has the desired zeros by construction but uses precisely the data one wants to explain. The probability-CDF centering takes neither route: both the Volterra part and its rank-one correction are determined by p before any zero is known, while the modified Fredholm determinant of the resulting one operator recovers the complete centered characteristic function. The CDF term also retains the endpoint datum that a bare Volterra operator loses.

General semiseparable determinant theory supplies a broad reduction framework for kernels of this type [5]. What is specific here is the probability normalization that makes the scalar reduction close exactly on a characteristic function, together with a whole-line Schatten criterion that can be verified with explicit constants for the Riemann–Theta density. Compared with the Theta boundary formulation in [6], the present operator is bounded and compact, is constructed without selecting a zero, and records the entire zero divisor and its algebraic multiplicities in one regularized determinant. These are complementary features rather than a claim that one framework subsumes the other.

Two further distinctions locate the construction more precisely. Connes's spectral side is global and representation-theoretic: it is built from the idele-class action on the Adele-class space, critical zeros are missing spectral lines, and possible noncritical zeros enter as resonances [4]. No Adele space, group representation, or arithmetic trace formula is used here; the operator in (1.2) is a compact operator on $L^2(\mathbb{R})$ obtained directly from a positive probability density. Suzuki [10], in a different integral-operator approach, studies the Hankel-type family $\mathsf{K}_\zeta^{\omega,\nu}[t]$ on $L^2(-\infty, t)$. Its kernel is defined through a shifted ratio of completed zeta functions, and the quotient

$$\frac{\det(I + \mathsf{K}_\zeta^{\omega,\nu}[t])}{\det(I - \mathsf{K}_\zeta^{\omega,\nu}[t])}$$

enters a de Branges Hamiltonian obtained from associated integral equations. By contrast, the present construction uses neither de Branges spaces nor the parameters (ω, ν, t): a single fixed Hilbert–Schmidt operator $\mathcal{B}_\Theta$ is determined by the classical Theta density before any zero is chosen, and its one modified determinant is directly $\Xi(z)/\Xi(0)$. Thus the common Fredholm vocabulary conceals different operator data, determinant identities, and spectral objectives.

The principal results are as follows.

(i) For a positive density on a finite interval, the CDF is the unique rank-one scalar centering compatible with the local projection principle, and the regularized determinant of the resulting operator is the characteristic function with its linear phase removed; see Proposition 3.2 and Theorem 3.3.

(ii) A transparent whole-line admissibility condition gives Hilbert–Schmidt convergence of symmetric truncations and hence the same determinant formula on $\mathbb{R}$; see Theorem 4.4.

(iii) The Riemann–Theta density satisfies these assumptions. Explicit first-mode estimates give

$$\|\mathcal{B}_\Theta\|_{\mathcal{S}_2}^2 \le \frac{1+\Delta}{\pi d_*} < \infty, \tag{1.4}$$

where Δ and d_* are given in equations (5.4) and (5.9).

(iv) The resulting determinant is exactly

$$\det_2(I + z\mathcal{B}_\Theta) = \frac{\Xi(z)}{\Xi(0)}. \tag{1.5}$$

Thus the zero divisor, including algebraic multiplicities, is realized by a single compact operator.

(v) The same integral kernel is the Moore–Penrose inverse of a natural closed first-order Theta operator. This gives exact inverse identities and an explicit positive lower bound without using the invalid implication "injective implies bounded below." The first-order factor has a Witten-type supersymmetric completion and Fredholm index -1.

Up to an overall phase and interchange of the two partner sectors, the first-order factor is the one-dimensional Witten differential associated with the logarithmic potential $\Phi = -\log p$ [12]. The PT symmetry of the compact inverse belongs to the non-Hermitian framework initiated in [1], while the possible metric intertwining equation is naturally read in the pseudo-Hermitian framework of [8, 7]. These connections sharpen the spectral question but do not supply spectral reality.

The logical boundary is important. Equation (1.5) is an operator realization of the full zero divisor, not a proof that the divisor is real. In fact,

$$\mathrm{RH} \quad \Longleftrightarrow \quad \mathrm{Spec}(\mathcal{B}_\Theta) \setminus \{0\} \subset \mathbb{R}.$$

PT symmetry supplies the expected quartet symmetry, but PT-symmetric compact operators need not have real spectrum. The missing step is therefore a specific spectral-reality theorem for the arithmetic kernel, not a domain closure, determinant normalization, or numerical approximation.

The determinant and trace-ideal facts used below are standard; our conventions follow Simon [9]. Accordingly, the finite-interval calculation in Theorem 3.3 is presented as a self-contained scalar specialization, not as a claim that rank-one Volterra determinant reduction is new. The contribution here is the probability-CDF centering, the explicit full-line Schatten closure for the Riemann density, its Moore–Penrose interpretation, and the exact specialization to Ξ. Standard properties of ξ and its Fourier kernel may be found in Titchmarsh's account [11].

2. The positive Riemann–Theta density

For $x \geq 0$, set

$$W(x) = 2\sum_{n=1}^{\infty} \left(2\pi^2 n^4 e^{9x/2} - 3\pi n^2 e^{5x/2}\right) e^{-\pi n^2 e^{2x}}, \qquad W(-x) = W(x). \tag{2.1}$$

The Jacobi transformation law shows that this even extension is smooth. It is the logarithmic form of the classical Theta density appearing in the Mellin representation of ξ. For completeness, a derivation from the classical

Jacobi inversion and Mellin formulas, independent of any recent Theta-based spectral construction, is given in section A; compare [11].

Proposition 2.1 (Riemann Fourier representation)**.** *The function W is strictly positive, even, smooth, and has exponential moments of every order. Moreover,*

$$\Xi(z) = \int_{\mathbb{R}} W(x)e^{izx}\,\mathrm{d}x, \qquad z \in \mathbb{C}. \tag{2.2}$$

In particular,

$$c_0 := \int_{\mathbb{R}} W(x)\,\mathrm{d}x = \Xi(0) > 0. \tag{2.3}$$

Proof. The complete calculation is recorded in section A. In brief, if $\vartheta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}$ and

$$\omega(t) = \frac{1}{2}\big(2t^2\vartheta''(t) + 3t\vartheta'(t)\big),$$

then Jacobi inversion gives $\omega(1/t) = t^{1/2}\omega(t)$, and $W(x) = 2e^{x/2}\omega(e^{2x})$. This identity proves smoothness and evenness without gluing two half-line formulas. The series is positive for $x \geq 0$ because $2\pi n^2 e^{2x} - 3 > 0$, hence it is positive everywhere by evenness, and it has double-exponential decay. Two integrations by parts in the classical Mellin formula give

$$\xi(s) = \int_0^\infty \omega(t)t^{s/2}\,\frac{\mathrm{d}t}{t}.$$

The substitution $t = e^{2x}$ at $s = \frac{1}{2} + iz$ is exactly (2.2). Double-exponential decay gives absolute convergence locally uniformly in $z \in \mathbb{C}$. □

Henceforth put

$$p(x) = \frac{W(x)}{c_0}, \qquad F(x) = \int_{-\infty}^{x} p(y)\,\mathrm{d}y. \tag{2.4}$$

Then p is an even, strictly positive probability density, $F(-x) = 1 - F(x)$, and

$$\varphi_p(z) := \int_{\mathbb{R}} p(x)e^{izx}\,\mathrm{d}x = \frac{\Xi(z)}{\Xi(0)}. \tag{2.5}$$

3. The finite-interval determinant identity

We first isolate the operator-theoretic mechanism from the arithmetic specialization. Throughout, Hilbert-space inner products are linear in the first variable. For vectors u, v, the notation $u \otimes v$ means $(u \otimes v)f = \langle f, v\rangle u$.

Let $-\infty < a < b < \infty$, and let p be a probability density on $[a, b]$ that is bounded above and bounded away from zero. Define

$$F(x) = \int_a^x p(y)\,\mathrm{d}y, \qquad \mu = \int_a^b xp(x)\,\mathrm{d}x. \tag{3.1}$$

Definition 3.1 (Centered Volterra operator)**.** *The centered Volterra operator associated with p is*

$$(\mathfrak{V}_p f)(x) = \frac{i}{\sqrt{p(x)}} \left[\int_a^x \sqrt{p(y)} f(y)\, \mathrm{d}y - F(x) \int_a^b \sqrt{p(y)} f(y)\, \mathrm{d}y \right]. \tag{3.2}$$

The particular correction F in (3.2) is canonical within a natural class. Put $P_p = I - \sqrt{p} \otimes \sqrt{p}$. For an absolutely continuous function h with $h(a) = 0$, define

$$(\mathcal{B}_{p,h} f)(x) = \frac{i}{\sqrt{p(x)}} \left[\int_a^x \sqrt{p(y)} f(y)\, \mathrm{d}y - h(x) \int_a^b \sqrt{p(y)} f(y)\, \mathrm{d}y \right]. \tag{3.3}$$

Proposition 3.2 (Uniqueness of CDF centering)**.** *The local projection identity*

$$\frac{d}{dx} \left(\frac{\sqrt{p}\, \mathcal{B}_{p,h} f}{i} \right) = \sqrt{p}\, P_p f \qquad \text{for every } f \in L^2([a, b]) \tag{3.4}$$

holds in the weak sense if and only if $h = F$. For this unique choice, the bracket in (3.3) vanishes at both a and b for every f.

Proof. Writing $m_f = \langle f, \sqrt{p} \rangle$, the two sides of (3.4) are respectively

$$\sqrt{p}\, f - h' m_f \qquad \text{and} \qquad \sqrt{p}\, f - p m_f.$$

Taking $f = \sqrt{p}$ shows that the identity holds for every f only if $h' = p$ almost everywhere. Since $h(a) = 0$, this is equivalent to $h = F$; the converse is immediate. Finally, $F(a) = 0$ and $F(b) = 1$, so the bracket vanishes at both endpoints. $\square$

The operator $\mathfrak{V}_p$ in Definition 3.1 is Hilbert–Schmidt. Conjugation by multiplication with $\sqrt{p}$ turns it into

$$C = i(\mathcal{V} - F \otimes \mathbf{1}), \qquad (\mathcal{V} f)(x) = \int_a^x f(y)\, \mathrm{d}y, \tag{3.5}$$

on $L^2([a, b])$. The multiplication operators are bounded and boundedly invertible, so the regularized determinant is unchanged by this similarity.

For $K \in \mathcal{S}_2$, we use

$$\det{}_2(I + K) = \det\big((I + K)e^{-K}\big). \tag{3.6}$$

It is an entire function of scalar multiples of K and is continuous in the Hilbert–Schmidt norm [9, Chapter 9]. We use the standard Hilbert–Schmidt product formula in the form recorded, for example, in [3].

Theorem 3.3 (Finite centered-Volterra determinant)**.** *For every $z \in \mathbb{C}$,*

$$\boxed{\det{}_2(I + z\mathfrak{V}_p) = e^{-i\mu z} \int_a^b p(x) e^{izx}\, \mathrm{d}x.} \tag{3.7}$$

Proof. The Volterra operator $\mathcal{V}$ is Hilbert–Schmidt and quasinilpotent; hence

$$\det_2(I + iz\mathcal{V}) = 1. \tag{3.8}$$

Put

$$Q = (I + iz\mathcal{V})^{-1}F.$$

Since $\mathcal{V}$ commutes with its resolvent, the factorization

$$I + zC = (I + iz\mathcal{V})(I - izQ \otimes \mathbf{1}) \tag{3.9}$$

holds. The multiplicative formula for $\det_2$ gives

$$\begin{aligned}\det_2(I + zC) = \Big(1 - iz\int_a^b Q(x)\,\mathrm{d}x\Big)\exp\Big(iz\int_a^b Q(x)\,\mathrm{d}x \\ - z^2\int_a^b (\mathcal{V}Q)(x)\,\mathrm{d}x\Big).\end{aligned} \tag{3.10}$$

The resolvent identity $(I + iz\mathcal{V})Q = F$ shows that the exponent in (3.10) is $iz\int_a^b F$.

For the scalar factor, set

$$q = (I + iz\mathcal{V})^{-1}p.$$

Because $F = \mathcal{V}p$, one has $Q = \mathcal{V}q$. Therefore

$$1 - iz\int_a^b Q = \int_a^b q = Q(b). \tag{3.11}$$

The function Q satisfies the initial-value problem

$$Q'(x) + izQ(x) = p(x), \qquad Q(a) = 0, \tag{3.12}$$

and consequently

$$Q(b) = e^{-izb}\int_a^b p(x)e^{izx}\,\mathrm{d}x. \tag{3.13}$$

Finally, Fubini's theorem gives

$$\int_a^b F(x)\,\mathrm{d}x = \int_a^b (b-y)p(y)\,\mathrm{d}y = b - \mu. \tag{3.14}$$

Combining (3.10)–(3.14) proves (3.7). Similarity in (3.5) returns the result to $\mathfrak{V}_p$. □

Corollary 3.4 (Finite trace–cumulant identity)**.** *Let $\kappa_n(p)$ be the cumulants of p. For every $n \geq 2$,*

$$\operatorname{tr}(\mathfrak{V}_p^n) = \frac{(-1)^{n+1}i^n}{(n-1)!}\,\kappa_n(p). \tag{3.15}$$

Proof. Near $z = 0$, compare

$$\log\det_2(I + z\mathfrak{V}_p) = \sum_{n=2}^{\infty}\frac{(-1)^{n+1}}{n}z^n\operatorname{tr}(\mathfrak{V}_p^n)$$

with the cumulant expansion of $e^{-i\mu z}\int p(x)e^{izx}\,\mathrm{d}x$. □

4. Whole-line admissibility and closure

The finite-interval formula does not pass automatically to $\mathbb{R}$: the centered Volterra kernel may fail to be Hilbert–Schmidt, even for familiar density functions. We now state a directly checkable class for which the passage is valid.

Let p be a strictly positive, continuous, even probability density on $\mathbb{R}$, and let $F(x) = \int_{-\infty}^{x} p$. Define formally

$$(\mathfrak{V}_p f)(x) = \frac{i}{\sqrt{p(x)}} \left[\int_{-\infty}^{x} \sqrt{p(y)} f(y)\,\mathrm{d}y - F(x) \int_{\mathbb{R}} \sqrt{p(y)} f(y)\,\mathrm{d}y \right]. \tag{4.1}$$

Definition 4.1 (Volterra-admissible density)**.** *An even density p is called* Volterra-admissible *if:*

(A1) $p(x) > 0$ for all x and $\int_{\mathbb{R}} p(x) e^{M|x|}\,\mathrm{d}x < \infty$ for every $M > 0$;
(A2)

$$I_p := \int_{\mathbb{R}} \frac{F(x)(1 - F(x))}{p(x)}\,\mathrm{d}x < \infty; \tag{4.2}$$

(A3) if $q_R = \int_{R}^{\infty} p(x)\,\mathrm{d}x$, then

$$\frac{q_R^2}{1 - 2q_R} \int_{-R}^{R} \frac{(2F(x) - 1)^2}{p(x)}\,\mathrm{d}x \longrightarrow 0. \tag{4.3}$$

The terminology is local to this paper. Condition (4.2) is exactly the Hilbert–Schmidt requirement; condition (4.3) controls the normalization error created by symmetric truncation.

Proposition 4.2 (Exact Hilbert–Schmidt norm)**.** *If (4.2) holds, then (4.1) defines an operator in $\mathcal{S}_2(L^2(\mathbb{R}))$ with kernel*

$$b_p(x, y) = i \sqrt{\frac{p(y)}{p(x)}} \left(\mathbf{1}_{\{y \le x\}} - F(x)\right) \tag{4.4}$$

and

$$\boxed{\|\mathfrak{V}_p\|_{\mathcal{S}_2}^2 = \int_{\mathbb{R}} \frac{F(x)(1 - F(x))}{p(x)}\,\mathrm{d}x.} \tag{4.5}$$

Proof. Tonelli's theorem applies to $|b_p|^2$. For fixed x,

$$\begin{aligned} \int_{\mathbb{R}} |b_p(x, y)|^2\,\mathrm{d}y &= \frac{1}{p(x)} \left[(1 - F(x))^2 \int_{-\infty}^{x} p(y)\,\mathrm{d}y + F(x)^2 \int_{x}^{\infty} p(y)\,\mathrm{d}y \right] \\ &= \frac{F(x)(1 - F(x))}{p(x)}. \end{aligned}$$

Integrating the preceding identity over all $x \in \mathbb{R}$ gives (4.5) and proves the claim. ☐

For $R > 0$, define

$$p_R(x) = \frac{p(x)\mathbf{1}_{[-R,R]}(x)}{1 - 2q_R}, \qquad F_R(x) = \int_{-R}^{x} p_R(y)\,\mathrm{d}y, \tag{4.6}$$

and let $\mathfrak{V}_p^{(R)}$ be the finite-interval operator, extended by zero to $L^2(\mathbb{R})$.

Proposition 4.3 (Hilbert–Schmidt truncation)**.** *If p is Volterra-admissible, then*

$$\|\mathfrak{V}_p^{(R)} - \mathfrak{V}_p\|_{\mathcal{S}_2} \longrightarrow 0. \tag{4.7}$$

Proof. On $[-R, R]$ one has the exact identity

$$F_R(x) - F(x) = \frac{q_R(2F(x) - 1)}{1 - 2q_R}. \tag{4.8}$$

The normalization in p_R cancels from the ratio $\sqrt{p_R(y)/p_R(x)}$. Therefore the squared kernel error on $[-R, R]^2$ is exactly

$$E_R^{\text{in}} = \frac{q_R^2}{1 - 2q_R} \int_{-R}^{R} \frac{(2F(x) - 1)^2}{p(x)} \,\mathrm{d}x, \tag{4.9}$$

which tends to zero by (4.3). On the complement of the square, the truncated kernel vanishes. The remaining error is the integral of $|b_p|^2$ over a decreasing family of sets with empty intersection; it tends to zero by Proposition 4.2 and continuity from above. □

Theorem 4.4 (Whole-line determinant)**.** *If p is Volterra-admissible, then*

$$\boxed{\det_2(I + z\mathfrak{V}_p) = \int_{\mathbb{R}} p(x) e^{izx} \,\mathrm{d}x, \qquad z \in \mathbb{C}.} \tag{4.10}$$

The convergence of the finite determinants to both sides is locally uniform on $\mathbb{C}$.

Proof. Each p_R is even, so its mean is zero. By Theorem 3.3,

$$\det_2(I + z\mathfrak{V}_p^{(R)}) = \int_{-R}^{R} p_R(x) e^{izx} \,\mathrm{d}x.$$

The left side converges locally uniformly by Proposition 4.3 and continuity estimates for $\det_2$ on $\mathcal{S}_2$. For $|\operatorname{Im} z| \leq M$, the right side converges uniformly by condition (A1), normalization of p_R, and dominated convergence. The two limits agree. □

Remark 4.5 (Nonzero mean)**.** *For a non-even density of mean μ, the same argument, with two-sided truncation defects written separately, gives*

$$\det_2(I + z\mathfrak{V}_p) = e^{-i\mu z} \varphi_p(z).$$

Thus the regularization removes exactly the first cumulant.

Example 4.6. *The densities $p_r(x) = c_r e^{-|x|^r}$ are Volterra-admissible for $r > 2$. Indeed, their tail-to-density ratio is $O(x^{1-r})$, which is integrable precisely beyond the Gaussian exponent. At $r = 2$, the Gaussian density fails (4.2): Mills' ratio gives $F(x)(1 - F(x))/p(x) \sim (2x)^{-1}$ as $x \to +\infty$. Hence whole-line Hilbert–Schmidt closure is a genuine restriction, not a formal consequence of rapid decay.*

Remark 4.7 (The reciprocal log-slope test)**.** *The Gaussian obstruction is not a lack of moments: the Gaussian has exponential moments of every order. The obstruction comes from the $p^{-1/2}$ normalization in the kernel. On the positive tail, $F(x) \to 1$, so the Hilbert–Schmidt integrand is asymptotic to $(1-F(x))/p(x)$. For a smooth, eventually log-convex tail $p(x) = c\exp(-\Phi(x))$ satisfying $\Phi'(x) \to \infty$ and $\Phi''(x)/(\Phi'(x))^2 \to 0$, endpoint Laplace estimates give*

$$\frac{1-F(x)}{p(x)} \sim \frac{1}{\Phi'(x)}. \tag{4.11}$$

Thus admissibility tests the integrability of the reciprocal logarithmic slope, not merely rapid decay. For $\Phi(x) = x^2$, the right side is $(2x)^{-1}$ and diverges logarithmically. By contrast, equations (5.2) and (5.5) show that the Riemann–Theta density has

$$-\log p(x) = \pi e^{2x} - \frac{9}{2}x + O(1), \qquad x \to +\infty, \tag{4.12}$$

so its reciprocal log-slope is of order e^{-2x}. The rigorous version is the tail ratio bound (5.8). This double-exponential geometry is exactly what separates the Theta density from the Gaussian threshold.

5. Explicit Theta-tail estimates

We now verify Definition 4.1 for the density in equation (2.4) and derive a fixed-constant norm bound.

For $x \geq 0$, put

$$q = \pi e^{2x} \geq \pi \tag{5.1}$$

and denote the $n = 1$ summand of (2.1) by

$$w_1(x) = 2\pi^{-1/4} q^{5/4}(2q-3)e^{-q}. \tag{5.2}$$

For $n \geq 2$, the ratio of the nth summand to w_1 is

$$\frac{w_n(x)}{w_1(x)} = n^2 \frac{2qn^2-3}{2q-3} e^{-q(n^2-1)} \leq 2n^4 e^{-q(n^2-1)}. \tag{5.3}$$

Since $n^2 - 1 \geq 3(n-1)$, define

$$r = e^{-3\pi}, \qquad \Delta = 2\left[\frac{1+11r+11r^2+r^3}{(1-r)^5} - 1\right]. \tag{5.4}$$

Summing (5.3) gives

$$\boxed{w_1(x) \leq W(x) \leq (1+\Delta)w_1(x), \qquad x \geq 0.} \tag{5.5}$$

We use the elementary incomplete-Gamma estimate

$$\int_q^\infty t^a e^{-t}\,\mathrm{d}t \leq \frac{q^a e^{-q}}{1-a/q}, \qquad q > a \geq 0. \tag{5.6}$$

It follows from $t^a \leq q^a \exp(a(t-q)/q)$ for $t \geq q$.

Let $T(x) = \int_x^\infty W(y)\,\mathrm{d}y$. The substitution $t = \pi e^{2y}$ yields

$$\int_x^\infty w_1(y)\,\mathrm{d}y = \pi^{-1/4}\int_q^\infty t^{1/4}(2t-3)e^{-t}\,\mathrm{d}t. \tag{5.7}$$

Combining (5.5)–(5.7) gives

$$\frac{T(x)}{W(x)} \le \frac{1+\Delta}{(2q-3)(1-5/(4q))} \le \frac{1+\Delta}{\pi d_*}e^{-2x}, \tag{5.8}$$

where

$$d_* = \left(2-\frac{3}{\pi}\right)\left(1-\frac{5}{4\pi}\right) > 0. \tag{5.9}$$

Theorem 5.1 (Theta admissibility and norm bound)**.** *The density $p = W/c_0$ is Volterra-admissible. The corresponding operator $\mathcal{B}_\Theta := \mathfrak{V}_p$ satisfies*

$$\boxed{\|\mathcal{B}_\Theta\|_{\mathcal{S}_2}^2 \le \frac{1+\Delta}{\pi d_*}.} \tag{5.10}$$

Proof. For $x \ge 0$, $c_0(1-F(x)) = T(x)$ and $F(x) \le 1$. By evenness and (5.8),

$$\begin{aligned}\|\mathcal{B}_\Theta\|_{\mathcal{S}_2}^2 &= c_0\int_{\mathbb{R}}\frac{F(x)(1-F(x))}{W(x)}\,\mathrm{d}x\\ &\le 2\int_0^\infty \frac{T(x)}{W(x)}\,\mathrm{d}x \le \frac{1+\Delta}{\pi d_*}.\end{aligned} \tag{5.11}$$

This proves condition (A2).

It remains to control symmetric truncation. Put

$$Q_R = \pi e^{2R}, \qquad C_q = \frac{2(1+\Delta)\pi^{-1/4}}{c_0(1-5/(4\pi))}. \tag{5.12}$$

Equations (5.5) and (5.6) imply

$$q_R \le C_q Q_R^{5/4}e^{-Q_R}. \tag{5.13}$$

The lower bound $W \ge w_1$ gives

$$\int_{-R}^{R}\frac{\mathrm{d}x}{p(x)} \le \frac{c_0}{2\pi^2(2\pi-3)}e^{Q_R}. \tag{5.14}$$

For all sufficiently large R, $q_R \le 1/4$; using $(2F-1)^2 \le 1$, the expression in (4.3) is at most

$$\frac{C_q^2 c_0}{\pi^2(2\pi-3)}Q_R^{5/2}e^{-Q_R}, \tag{5.15}$$

which tends to zero. This proves condition (A3).

Finally, inserting $e^{M|x|}$ in the same calculation, with $a_M = 5/4 + M/2$ and $Q_R > a_M$, gives

$$\int_{|x|>R} p(x)e^{M|x|}\,\mathrm{d}x \le \frac{4(1+\Delta)\pi^{-1/4-M/2}}{c_0(1-a_M/Q_R)}Q_R^{a_M}e^{-Q_R}. \tag{5.16}$$

Thus condition (A1) also holds. □

Corollary 5.2 (Fredholm realization of Ξ)**.** *For every* $z \in \mathbb{C}$*,*

$$\boxed{\det_2(I + z\mathcal{B}_\Theta) = \frac{\Xi(z)}{\Xi(0)}.} \tag{5.17}$$

Proof. Combine equation (2.5) and Theorems 4.4 and 5.1. □

6. A first-order operator and its Moore–Penrose inverse

The integral operator also has a differential interpretation. Let

$$\psi = \sqrt{p}, \qquad P_0 = I - \psi \otimes \psi. \tag{6.1}$$

Since p is normalized, $\|\psi\|_2 = 1$. Put

$$D = -i\frac{d}{dx} \tag{6.2}$$

and define the maximal realization

$$\mathrm{Dom}(A) = \{g \in L^2(\mathbb{R}) : u = \sqrt{p}\, g \in H^1(\mathbb{R}),\ p^{-1/2}Du \in L^2(\mathbb{R})\},$$
$$Ag = p^{-1/2}D(\sqrt{p}\, g). \tag{6.3}$$

The space $C_c^\infty(\mathbb{R})$ is contained in $\mathrm{Dom}(A)$, so A is densely defined.

Proposition 6.1. *The operator* A *is closed. If* $\Phi = -\log p$ *and* $V = \frac{1}{2}\Phi'$*, then on* $C_c^\infty(\mathbb{R})$*,*

$$A = D + iV. \tag{6.4}$$

Proof. Suppose $g_n \to g$ and $Ag_n \to h$ in L^2. Since $\sqrt{p}$ is bounded, $\sqrt{p}\, g_n \to \sqrt{p}\, g$ and $D(\sqrt{p}\, g_n) = \sqrt{p}\, Ag_n \to \sqrt{p}\, h$ in L^2. Closedness of D on $H^1(\mathbb{R})$ gives $g \in \mathrm{Dom}(A)$ and $Ag = h$. Formula (6.4) is a direct product-rule calculation. □

Theorem 6.2 (Explicit Moore–Penrose inverse)**.** *The following identities hold:*

$$A\mathcal{B}_\Theta = P_0 \qquad \textit{on } L^2(\mathbb{R}), \tag{6.5}$$
$$\mathcal{B}_\Theta A = I \qquad \textit{on } \mathrm{Dom}(A), \tag{6.6}$$
$$\mathcal{B}_\Theta \psi = 0. \tag{6.7}$$

Consequently,

$$\boxed{\mathcal{B}_\Theta = A^\dagger,} \tag{6.8}$$

A *is injective,* $\mathrm{Ran}(A) = \psi^\perp$ *is closed, and*

$$\boxed{\|Ag\|_2 \geq \sqrt{\frac{\pi d_*}{1+\Delta}}\, \|g\|_2, \qquad g \in \mathrm{Dom}(A).} \tag{6.9}$$

Proof. For $f \in L^2(\mathbb{R})$, set

$$m = \int_{\mathbb{R}} \sqrt{p(y)}\, f(y)\, \mathrm{d}y, \qquad q_f(x) = \int_{-\infty}^{x} \sqrt{p(y)}\, f(y)\, \mathrm{d}y - F(x)m. \tag{6.10}$$

Then $\sqrt{p}\,\mathcal{B}_\Theta f = iq_f$ and

$$q_f'(x) = \sqrt{p(x)}\,(P_0 f)(x). \tag{6.11}$$

The Hilbert–Schmidt bound shows that $\mathcal{B}_\Theta f \in L^2$; hence $q_f \in L^2$. Equation (6.11) puts q_f in H^1 and gives $A\mathcal{B}_\Theta f = P_0 f$.

Conversely, let $g \in \mathrm{Dom}(A)$ and put $u = \sqrt{p}\, g$. Every $H^1(\mathbb{R})$ function has a continuous representative tending to zero at both endpoints. If $f = Ag$, then

$$\sqrt{p}\, f = Du = -iu'. \tag{6.12}$$

Moreover, $\sqrt{p}\, f \in L^1$ by Cauchy–Schwarz. Therefore

$$\int_{-\infty}^{x} \sqrt{p}\, f = -iu(x), \qquad \int_{\mathbb{R}} \sqrt{p}\, f = 0. \tag{6.13}$$

Substitution in (1.2) gives $\mathcal{B}_\Theta A g = g$. Taking $f = \psi$ in (6.10) gives $q_f = 0$, hence $\mathcal{B}_\Theta \psi = 0$.

The three identities imply that A is injective and has range exactly $\psi^\perp$, while $\mathcal{B}_\Theta$ vanishes on the orthogonal complement of the range. These are the Moore–Penrose equations, proving (6.8). Finally,

$$\|g\|_2 = \|\mathcal{B}_\Theta A g\|_2 \le \|\mathcal{B}_\Theta\|\, \|Ag\|_2 \le \|\mathcal{B}_\Theta\|_{\mathcal{S}_2}\, \|Ag\|_2,$$

and (6.9) follows from (5.10). □

Corollary 6.3 (Witten-type factorization and index)**.** *Let $q = iA$ and define, on the natural block domain,*

$$\mathcal{Q} = \begin{pmatrix} 0 & q^* \\ q & 0 \end{pmatrix}, \qquad \Gamma = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}. \tag{6.14}$$

Then $\mathcal{Q}$ is self-adjoint, $\Gamma\mathcal{Q} = -\mathcal{Q}\Gamma$, and

$$\mathcal{Q}^2 = \begin{pmatrix} A^*A & 0 \\ 0 & AA^* \end{pmatrix}. \tag{6.15}$$

On $C_c^\infty(\mathbb{R})$,

$$\begin{aligned} q &= \frac{d}{dx} - V, & q^* &= -\frac{d}{dx} - V, \\ q^*q &= -\frac{d^2}{dx^2} + V^2 + V', & qq^* &= -\frac{d^2}{dx^2} + V^2 - V'. \end{aligned} \tag{6.16}$$

Moreover,

$$\mathrm{Ker}(q) = \{0\}, \qquad \mathrm{Ker}(q^*) = \mathrm{span}\{\psi\}, \qquad \mathrm{ind}\, A = \mathrm{ind}\, q = -1. \tag{6.17}$$

Proof. The closed densely defined operator $q = iA$ produces the standard self-adjoint block supercharge in (6.14); its square and grading relation are immediate. Formula (6.16) follows from (6.4). By Theorem 6.2, $\operatorname{Ker}(A) = \{0\}$ and $\operatorname{Ran}(A) = \psi^{\perp}$ is closed. Hence $\operatorname{Ker}(A^*) = \operatorname{Ran}(A)^{\perp} = \operatorname{span}\{\psi\}$, which proves (6.17). ☐

Up to an overall phase and the conventional interchange of partner sectors, equations (6.14) and (6.16) are the one-dimensional Witten supersymmetric factorization with Witten function $-\Phi/2$ (equivalently, superpotential derivative $-V$) [12]. In this language $\psi = \sqrt{p}$ is the unique zero mode in one partner sector, while $\mathcal{B}_\Theta = A^\dagger$ is the bounded Moore–Penrose inverse of the first-order factor on $\psi^\perp$. It is not a resolvent of either self-adjoint partner Hamiltonian, and supersymmetry alone does not imply that the nonnormal compact operator $\mathcal{B}_\Theta$ has real spectrum.

Remark 6.4. *The order of proof matters. Injectivity of a closed operator does not imply a positive lower bound. Here the valid chain is*

$$\begin{aligned} \textit{explicit tail estimate} &\Longrightarrow \mathcal{B}_\Theta \in \mathcal{S}_2 \\ &\Longrightarrow (A\mathcal{B}_\Theta = P_0,\ \mathcal{B}_\Theta A = I) \\ &\Longrightarrow |A| \geq \|\mathcal{B}_\Theta\|^{-1} I. \end{aligned}$$

7. PT symmetry and the zero spectrum

Let parity P and conjugation T act by

$$(\mathsf{P}f)(x) = f(-x), \qquad (\mathsf{T}f)(x) = \overline{f(x)}. \tag{7.1}$$

Because p is even, V in (6.4) is real and odd.

Proposition 7.1 (Symmetries)**.** *On their natural domains,*

$$\mathsf{P}A\mathsf{P} = -A, \qquad \mathsf{T}A\mathsf{T} = -A, \qquad (\mathsf{PT})A(\mathsf{PT}) = A, \tag{7.2}$$

$$\mathsf{P}\mathcal{B}_\Theta\mathsf{P} = -\mathcal{B}_\Theta, \qquad (\mathsf{PT})\mathcal{B}_\Theta(\mathsf{PT}) = \mathcal{B}_\Theta. \tag{7.3}$$

Thus $\mathcal{B}_\Theta$ is parity-odd and PT-symmetric.

Proof. The relations for $A = D + iV$ follow from $\mathsf{P}DP = -D$, $\mathsf{T}D\mathsf{T} = -D$, and the oddness of V. The relations for $\mathcal{B}_\Theta = A^\dagger$ follow by applying the same unitary and antiunitary symmetries to the Moore–Penrose equations. They can also be read directly from (4.4) and $F(-x) = 1 - F(x)$. ☐

This is the standard antilinear symmetry setting of PT-symmetric quantum theory [1]. The distinction between symmetry and spectral reality is essential: PT symmetry permits real eigenvalues but does not force them. In the pseudo-Hermitian formulation [8, 7], one instead seeks a metric operator G satisfying $G\mathcal{B}_\Theta = \mathcal{B}_\Theta^* G$. If G is bounded, positive, and boundedly invertible, then $G^{1/2}\mathcal{B}_\Theta G^{-1/2}$ is self-adjoint, so the spectrum is real. Conversely, abstract metric reconstructions require additional diagonalizability and basis hypotheses; they cannot be imported here from PT symmetry alone.

We now state the exact spectral consequence of the determinant formula.

Theorem 7.2 (Reciprocal zero spectrum)**.** *The operator $\mathcal{B}_\Theta$ is compact and*

$$\boxed{\operatorname{Spec}(\mathcal{B}_\Theta) \setminus \{0\} = \left\{ -\frac{1}{\alpha} : \ \Xi(\alpha) = 0 \right\}.} \tag{7.4}$$

Algebraic multiplicities agree: the order of a zero α of Ξ equals the algebraic multiplicity of $-1/\alpha$ as an eigenvalue of $\mathcal{B}_\Theta$. Moreover,

$$\boxed{\text{RH} \iff \operatorname{Spec}(\mathcal{B}_\Theta) \setminus \{0\} \subset \mathbb{R}.} \tag{7.5}$$

Proof. Hilbert–Schmidt operators are compact. For a compact operator K, the zeros of $\det_2(I + zK)$ are precisely the points $z = -1/\lambda$, where $\lambda \neq 0$ is an eigenvalue of K; the zero order is the algebraic multiplicity [9, Chapter 9]. Apply this fact to Corollary 5.2. Finally, a nontrivial zeta zero $\rho = \beta + i\gamma$ corresponds to

$$\alpha = \gamma - i\left(\beta - \frac{1}{2}\right), \qquad \rho = \frac{1}{2} + i\alpha. \tag{7.6}$$

Thus $\beta = 1/2$ if and only if α, and hence $-1/\alpha$, is real. □

The symmetry relations imply that each nonzero spectral point occurs in the quartet

$$\lambda, \quad -\lambda, \quad \overline{\lambda}, \quad -\overline{\lambda}, \tag{7.7}$$

with collapses on the axes. This is the operator form of the evenness, reality, and functional-equation symmetries of Ξ.

The connection with the Theta boundary state can also be written directly. For $\alpha \in \mathbb{C}$, define

$$u_\alpha(x) = e^{-i\alpha x} \int_{-\infty}^{x} e^{i\alpha y} W(y)\, \mathrm{d}y, \qquad g_\alpha = \frac{u_\alpha}{\sqrt{W}}. \tag{7.8}$$

Lemma 7.3 (Boundary-state domain lemma)**.** *If $\Xi(\alpha) = 0$, then $g_\alpha \in \operatorname{Dom}(A)$. More precisely, there are constants $C_\alpha, R_\alpha > 0$ such that*

$$|u_\alpha(x)| \le C_\alpha e^{-2|x|} W(x), \qquad |x| \ge R_\alpha. \tag{7.9}$$

Proof. The zero condition and equation (2.2) give the second representation

$$u_\alpha(x) = -e^{-i\alpha x} \int_{x}^{\infty} e^{i\alpha y} W(y)\, \mathrm{d}y.$$

Use this formula as $x \to +\infty$ and the original formula as $x \to -\infty$. After taking absolute values, the only additional factor relative to (5.7) is $e^{|\Im\alpha|\,|y-x|}$. Under the substitution $t = \pi e^{2|y|}$ it changes the power of t by $|\Im\alpha|/2$. The incomplete-Gamma estimate (5.6), together with (5.5), therefore gives (7.9); the factor $e^{-2|x|}$ is the reciprocal of the leading scale $\pi e^{2|x|}$. In particular,

$$\int_{\mathbb{R}} \frac{|u_\alpha(x)|^2}{W(x)}\, \mathrm{d}x < \infty.$$

Moreover,

$$Du_\alpha = -\alpha u_\alpha - iW. \tag{7.10}$$

The same estimate shows that $u_\alpha, u'_\alpha \in L^2(\mathbb{R})$ and that $W^{-1/2}Du_\alpha \in L^2(\mathbb{R})$. Since $\sqrt{p}\, g_\alpha = u_\alpha/\sqrt{c_0}$, these are exactly the two conditions in (6.3). □

Equation (7.10), followed by integration over $\mathbb{R}$, gives $\int_{\mathbb{R}} u_\alpha = -ic_0/\alpha$; here $\alpha \neq 0$ because $\Xi(0) > 0$. It follows that

$$(A + \alpha P_0)g_\alpha = 0. \tag{7.11}$$

Applying $\mathcal{B}_\Theta$ to (7.11) and using equations (6.6) and (6.7) then gives

$$\mathcal{B}_\Theta g_\alpha = -\frac{1}{\alpha} g_\alpha. \tag{7.12}$$

This is the first-order, rank-one form of the Theta boundary mechanism in [6]; the determinant argument above additionally records all algebraic multiplicities without requiring a separate root-chain analysis.

8. Trace identities and structural consequences

The whole-line determinant theorem immediately extends Corollary 3.4.

Theorem 8.1 (Whole-line trace–cumulant calculus)**.** *Let p be Volterra-admissible, and let $\kappa_n(p)$ be its cumulants. Then, for every $n \geq 2$,*

$$\boxed{\operatorname{tr}(\mathfrak{V}_p^n) = \frac{(-1)^{n+1} i^n}{(n-1)!}\, \kappa_n(p).} \tag{8.1}$$

For the even Riemann density, all odd cumulants vanish. In particular,

$$\operatorname{tr}(\mathcal{B}_\Theta^2) = \operatorname{Var}_p(X) = -\frac{\Xi''(0)}{\Xi(0)}. \tag{8.2}$$

The same identity follows directly from the kernel. If $y < x$, then

$$b_p(x, y) b_p(y, x) = F(y)(1 - F(x)),$$

and integration over the two half-triangles gives the variance. This is an independent normalization check: no extra factor e^{az^2} can be inserted in (5.17) without changing (8.2).

There is also a useful conceptual separation. The Volterra indicator in (4.4) retains order; the CDF term retains the scalar endpoint memory removed by centering; only the final regularized determinant becomes a scalar. Thus the construction provides a general method for realizing an entire characteristic function as the scalar readout of an ordered compact channel. The Riemann case is distinguished by the arithmetic Theta density, not by the abstract determinant mechanism alone.

The construction is faithful modulo translation in a precise sense. For two densities to which the finite-interval theorem (or its admissible whole-line extension) applies, equality of their modified determinant functions is equivalent to equality of their centered probability laws. Indeed, Theorems 3.3 and 4.4, together with Remark 4.5, identify the determinants with the characteristic functions of $X - \mathbb{E}X$, and characteristic functions determine probability measures uniquely. Thus probability-CDF centering defines a canonical operator encoding of a law modulo its first cumulant; Theorem 8.1 recovers every higher cumulant as a trace.

9. Concluding remarks and the spectral-reality problem

The results above close five operator-theoretic points unconditionally:

(1) the CDF correction is uniquely forced by the local projection and endpoint requirements;
(2) $\mathcal{B}_\Theta$ is a well-defined Hilbert–Schmidt operator on the full real line, with an explicit bound;
(3) its finite centered restrictions converge in Hilbert–Schmidt norm;
(4) its regularized determinant is exactly $\Xi(z)/\Xi(0)$, including normalization and algebraic multiplicity;
(5) it is the Moore–Penrose inverse of a concrete closed first-order Theta operator, admits a Witten-type supersymmetric completion of index -1, and obeys exact PT and parity symmetries.

None of these statements implies that the spectrum is real. In the present language, the remaining RH problem is exactly the following.

Problem 9.1 (Theta spectral-reality problem)**.** *Prove, using a property specific to the arithmetic density W, that every nonzero eigenvalue of $\mathcal{B}_\Theta$ is real.*

Self-adjointness of $\mathcal{B}_\Theta$ in the ambient L^2 inner product is false in general, and PT symmetry by itself is insufficient. One possible direction is to seek a bounded, positive, invertible metric G satisfying $G\mathcal{B}_\Theta = \mathcal{B}_\Theta^* G$, as in pseudo-Hermitian quantum mechanics [8, 7]. Such a metric would make $\mathcal{B}_\Theta$ boundedly similar to a self-adjoint compact operator and hence force spectral reality. It is potentially stronger than RH alone: it would also exclude nontrivial Jordan chains and impose Riesz-basis control on the eigenvectors. Its existence should therefore not be treated as a free consequence of PT symmetry or of the rank-one construction. Another possible direction is to exploit additional total-positivity or variation-diminishing structure of the specific Theta kernel. The present paper does not assume either property.

The value of Problem 9.1 is that it is now separated from all regularization and closure issues. Any proposed RH argument using this model must act on the explicit kernel (4.4) and prove spectral reality without defining the missing structure from the zeros themselves.

Appendix A. Classical derivation of the Riemann–Theta kernel

This appendix derives (2.1) and (2.2) directly from the classical theta transformation and Mellin formula. Put

$$\vartheta(t) = \sum_{n\in\mathbb{Z}} e^{-\pi n^2 t}, \qquad \psi(t) = \frac{\vartheta(t)-1}{2} = \sum_{n=1}^{\infty} e^{-\pi n^2 t}, \qquad t>0. \tag{A.1}$$

Poisson summation gives Jacobi's inversion formula

$$\vartheta(1/t) = t^{1/2}\vartheta(t). \tag{A.2}$$

For $\Re s > 1$, termwise integration gives the classical Mellin identity

$$\pi^{-s/2}\Gamma(s/2)\zeta(s) = \int_0^\infty \psi(t)t^{s/2}\,\frac{\mathrm{d}t}{t}. \tag{A.3}$$

Define

$$\begin{aligned}\omega(t) &:= \frac{1}{2}\big(2t^2\vartheta''(t) + 3t\vartheta'(t)\big)\\ &= \sum_{n=1}^{\infty}(2\pi^2n^4t^2 - 3\pi n^2t)e^{-\pi n^2 t}.\end{aligned} \tag{A.4}$$

The differentiated theta series converges locally uniformly on $(0,\infty)$, so ω is smooth. Differentiating (A.2) once and twice yields

$$\begin{aligned}\vartheta'(1/t) &= -\frac{1}{2}t^{3/2}\vartheta(t) - t^{5/2}\vartheta'(t),\\ \vartheta''(1/t) &= \frac{3}{4}t^{5/2}\vartheta(t) + 3t^{7/2}\vartheta'(t) + t^{9/2}\vartheta''(t).\end{aligned} \tag{A.5}$$

Substitution in (A.4) gives the exact transformation law

$$\omega(1/t) = t^{1/2}\omega(t). \tag{A.6}$$

Let $a = s/2$. Since $\omega(t) = 2t^2\psi''(t) + 3t\psi'(t)$, two integrations by parts in (A.3) give, initially for $\Re s > 1$,

$$\begin{aligned}\int_0^\infty \omega(t)t^{a-1}\,\mathrm{d}t &= (1-2a)\int_0^\infty t^a\psi'(t)\,\mathrm{d}t\\ &= a(2a-1)\int_0^\infty \psi(t)t^{a-1}\,\mathrm{d}t\\ &= \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s) = \xi(s).\end{aligned} \tag{A.7}$$

The boundary terms vanish at infinity by exponential decay. At the origin, (A.2) gives

$$\psi(t) = \frac{1}{2}(t^{-1/2} - 1) + O\Big(t^{-1/2}e^{-\pi/t}\Big), \qquad t \downarrow 0,$$

which also makes the boundary terms vanish when $\Re s > 1$. Equation (A.6) shows that the left side of (A.7) converges locally uniformly for every $s \in \mathbb{C}$; hence (A.7) holds on $\mathbb{C}$ by analytic continuation.

Finally define, for every $x \in \mathbb{R}$,

$$W(x) = 2e^{x/2}\omega(e^{2x}). \tag{A.8}$$

For $x \geq 0$, inserting (A.4) into (A.8) is exactly the series (2.1). Moreover, (A.6) gives

$$W(-x) = 2e^{-x/2}\omega(e^{-2x}) = 2e^{x/2}\omega(e^{2x}) = W(x),$$

so the reflected definition in (2.1) is the restriction of one smooth even function. For $t \geq 1$, every summand of (A.4) is positive because $2\pi n^2 t - 3 > 0$; (A.6) extends positivity to $0 < t < 1$. The same series gives double-exponential decay of $W(x)$ as $x \to +\infty$, and evenness gives it as $x \to -\infty$. Thus W has exponential moments of every order. In (A.7), set $s = \frac{1}{2} + iz$ and substitute $t = e^{2x}$. Then

$$\begin{aligned}\Xi(z) &= \int_0^\infty \omega(t) t^{(1/2+iz)/2} \frac{\mathrm{d}t}{t} \\ &= \int_{-\infty}^\infty 2e^{x/2}\omega(e^{2x}) e^{izx}\,\mathrm{d}x = \int_{\mathbb{R}} W(x) e^{izx}\,\mathrm{d}x,\end{aligned}$$

which proves (2.2) with the normalization used throughout.

Appendix B. Details of the Theta truncation bounds

For completeness we record the two estimates used in Theorem 5.1. From (5.2), the substitution $t = \pi e^{2x}$ gives

$$\begin{aligned}\int_R^\infty w_1(x)\,\mathrm{d}x &= \pi^{-1/4} \int_{Q_R}^\infty t^{1/4}(2t-3)e^{-t}\,\mathrm{d}t \\ &\leq \frac{2\pi^{-1/4} Q_R^{5/4} e^{-Q_R}}{1 - 5/(4Q_R)}.\end{aligned} \tag{B.1}$$

Together with $W \leq (1+\Delta)w_1$, this proves (5.13).

For the reciprocal integral, $W \geq w_1$ gives

$$\begin{aligned}\int_{-R}^R \frac{\mathrm{d}x}{p(x)} &\leq 2c_0 \int_0^R \frac{\mathrm{d}x}{w_1(x)} \\ &= \frac{c_0 \pi^{1/4}}{2} \int_\pi^{Q_R} \frac{e^t}{t^{9/4}(2t-3)}\,\mathrm{d}t \\ &\leq \frac{c_0}{2\pi^2(2\pi - 3)} e^{Q_R},\end{aligned} \tag{B.2}$$

which is (5.14). Substitution in (4.9) gives (5.15). These fixed-constant inequalities are the analytic content behind the full-line closure; numerical tail fitting is not used.

Appendix C. Continuity of the regularized determinant

We recall the standard estimate that justifies the limiting passage. For $S, T \in \mathcal{S}_2$,

$$|\det{}_2(I+S) - \det{}_2(I+T)| \leq \|S-T\|_{\mathcal{S}_2} \exp\Big(C(1 + \|S\|_{\mathcal{S}_2} + \|T\|_{\mathcal{S}_2})^2\Big) \tag{C.1}$$

for an absolute constant C; see [9, Chapter 9]. On a compact z-set, apply this estimate to $S = z\mathfrak{V}_p^{(R)}$ and $T = z\mathfrak{V}_p$. The norms remain bounded by Proposition 4.3, proving locally uniform convergence.

Appendix D. Code and reproducibility

The complete reproducibility bundle is supplied with this manuscript as the ancillary archive `btheta-volterra-fredholm-reproducibility.zip`. It contains the LaTeX source and bibliography, the verification program `verify_btheta.py`, a pinned dependency file, and a README giving the exact commands. The program performs two independent numerical regression tests: it checks the Fourier normalization in equation (2.2) at high precision, and it applies midpoint Nystrom discretization to the finite-interval identity in Theorem 3.3; compare the general numerical framework in [2]. No external data are required. The source, verification program, and compiled manuscript are publicly available at https://github.com/AlexxxxxLiu/btheta-volterra-fredholm. The ancillary archive may also be distributed directly as supplementary material. These computations are not used in any proof. The analytic argument depends only on the displayed inequalities and standard trace-ideal continuity.

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